UNIT 8 · TOPIC 1(C)

Material Selection, Specification and Testing for Farm Equipment

Syllabus topic: Ferrous, non-ferrous, alloy and composite materials: composition, production techniques and selection for farm equipment

Match the material to the part's dominant duty

Part	Material
Plough share / point, tines, blades (wear + impact)	Medium/high-C or BORON steel, hardened to ~50–60 HRC, often with hard-facing at the edge (Kepner Ch. 7; Unit 6, Topics 3–4)
Mouldboard	3-layer / high-C steel, case-hardened wear face
Disc (plough/harrow)	Boron / high-C steel, heat-treated, sharpenable
Engine block, gear housing, pulleys	Grey / SG cast iron (damping, machinability)
Crankshaft, gears, axles (high-stress rotating)	Alloy steel (Cr–Ni), forged and heat-treated
Radiator, pistons, some housings	Aluminium alloys (light, conductive)
Bushes, bearings, radiator tubes	Copper alloys — bronze / brass
Tanks, guards, body panels, hoods	Mild-steel sheet or FRP composite

Specification and testing

- **Composition:** chemical analysis, spark/spectroscopy.
- **Mechanical properties:** tensile strength, yield, % elongation, impact (Charpy) toughness.
- **HARDNESS:** BRINELL (BHN — ball indenter), ROCKWELL (HRC — diamond cone), VICKERS (HV — diamond pyramid).
- BIS/IS standards specify materials for particular implements (shares, tines, discs).

Abrasion-resistant (AR / Hardox-type) plate grades

Grade	Brinell hardness	Typical use
AR200 / mild	BHN 170–250	Frames, body
AR235 (most common in farm equipment)	BHN ~220–260	General implements
AR400	BHN 360–440	Wear plates, tines
AR500	BHN 470–530	High-wear edges
AR600	BHN 560–625	Extreme abrasion

Heat treatment delivers the properties

Heat treatment is what actually produces the required hardness/toughness combination from the chosen steel:

- **Hardening + tempering** — high hardness with restored toughness.
- **Case-hardening / carburising** — hard, wear-resistant surface over a tough core.
- **Normalising, annealing** — refine grain, soften, relieve stress.

Practice MCQs

Q27. The main duty a plough share material must resist is:

(a) Electrical load
(b) Abrasive soil wear + impact
(c) Heat only
(d) Light

Q28. Target hardness for a hardened plough share/point is about:

(a) 20–25 HRC
(b) 35–40 HRC
(c) 50–60 HRC
(d) 90 HRC

Q31. The hardness test using a diamond pyramid indenter is:

(a) Brinell
(b) Rockwell
(c) Vickers
(d) Izod

Q32. The Rockwell C scale (HRC) uses a:

(a) Steel ball only
(b) Diamond cone indenter
(c) Flat punch
(d) Pendulum

Q33. Impact toughness of a material is measured by the:

(a) Tensile test
(b) Charpy/Izod impact test
(c) Hardness test
(d) Fatigue test

Q34. The process that actually develops the required hardness/toughness in a chosen steel is:

(a) Painting
(b) Heat treatment (hardening + tempering, case-hardening)
(c) Galvanising
(d) Machining

Q35. Case-hardening (carburising) gives a component:

(a) Soft surface, hard core
(b) Hard wear-resistant surface with a tough core
(c) Uniform softness
(d) Corrosion coating

Answers

Q27 — (b) Abrasive soil wear + impact
Q28 — (c) 50–60 HRC — *Shares are hardened to 50–60 HRC by heat treatment.*
Q31 — (c) Vickers
Q32 — (b) Diamond cone indenter
Q33 — (b) Charpy/Izod impact test
Q34 — (b) Heat treatment (hardening + tempering, case-hardening) — *Heat treatment develops the hardness/toughness of a chosen steel.*
Q35 — (b) Hard wear-resistant surface with a tough core — *Case-hardening: hard surface + tough core — ideal for wear parts.*

Sources: Standard materials-science and manufacturing texts (Callister; Kalpakjian; Rajput — Engineering Materials); BIS/IS material specifications; SSAB/industry data on tillage-tool steels (as given in the source notes, July 2026); Kepner Ch. 3 & 7 (materials for tillage tools); NPTEL materials lectures.